

TEST REPORT

Fantasy Tactics League– BIL482

Test Information

Project	Fantasy Tactics League
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Executive Summary

This document presents the test report for the Fantasy Football Management System developed as part of the BIL482 Software Quality course. The system is a modular, use-case driven backend application built with FastAPI and SQLite, implementing four core use cases: team building (UC1), points calculation (UC2), global leaderboard management (UC3), and gameweek data fetching (UC4).

The test suite consists of 34 test cases spanning unit tests, pattern-level tests, and live API integration tests, written using the pytest framework and organized per use case for full traceability. Tests ran against a locally hosted Uvicorn server with a populated SQLite database containing real Gameweek 31 data. All 34 test cases passed. Three bugs were identified and resolved during the testing phase.

Related Requirements

Req ID	Description	Use Case	Design Pattern
UC1-REQ-01	System shall accept a valid 15-player squad	UC1 Team Builder	Strategy, Factory
UC1-REQ-02	Total squad cost must not exceed budget limit	UC1 Team Builder	Strategy (BudgetValidation)
UC1-REQ-03	Squad must contain exactly 15 players	UC1 Team Builder	Strategy (SquadSizeValidation)
UC1-REQ-04	Position distribution: 2 GK, 5 DEF, 5 MID, 3 FWD	UC1 Team Builder	Strategy (PositionValidation)
UC1-REQ-05	Maximum 3 players from any one real club	UC1 Team Builder	Strategy (TeamLimitValidation)
UC2-REQ-01	GK earns points for clean sheets, saves, away CS bonus	UC2 Points Calculator	Strategy (GoalkeeperScoringStrategy)
UC2-REQ-02	MID playing 90 min with ICT ≥ 8 earns +2 bonus	UC2 Points Calculator	Strategy (MidfielderScoringStrategy)
UC2-REQ-03	FWD with ICT ≥ 10 but no goal/assist earns +3	UC2 Points Calculator	Strategy (ForwardScoringStrategy)
UC2-REQ-04	Captain's score is doubled	UC2 Points Calculator	Decorator (ModifierStrategy)
UC2-REQ-05	Bench players contribute 0 points	UC2 Points Calculator	Decorator (ModifierStrategy)
UC3-REQ-01	Only one GlobalLeaderboard instance in memory	UC3 Leaderboard	Singleton
UC3-REQ-02	Leaderboard updates must trigger Observer notifications	UC3 Leaderboard	Observer
UC3-REQ-03	Leaderboard must be sorted by score descending	UC3 Leaderboard	Singleton
UC4-REQ-01	GET /players must return all players from the database	UC4 Data Fetcher	REST API

Req ID	Description	Use Case	Design Pattern
UC4-REQ-02	GET /stats/latest must resolve to the current gameweek	UC4 Data Fetcher	REST API
UC4-REQ-03	sort parameter must be sanitized to prevent SQL injection	UC4 Data Fetcher	REST API
UC4-REQ-04	GET /player/{id}/history must return 404 for unknown IDs	UC4 Data Fetcher	REST API

Test Cases & Results

UC1 – Team Builder

TC ID	Test Case	Input	Expected	Actual	Result
TC-01	Valid squad within budget	15 players, cost=75m	True, 'Valid budget'	True	☒ PASS
TC-02	Squad over budget	15 players, cost=150m	False, 'Budget exceeded'	False	☒ PASS
TC-03	Budget at exact limit	15 players, cost=100m	True	True	☒ PASS
TC-04	Only 14 players	14 player objects	False, 'Invalid squad size: 14'	False	☒ PASS
TC-05	16 players	16 player objects	False, 'Invalid squad size: 16'	False	☒ PASS
TC-06	Valid position distribution	2 GK, 5 DEF, 5 MID, 3 FWD	True	True	☒ PASS
TC-07	Wrong GK count (3 GKs)	3 GK, 4 DEF, 5 MID, 3 FWD	False, 'Goalkeeper'	False	☒ PASS
TC-08	Club limit respected	≤3 per club	True	True	☒ PASS

UC2 – Points Calculator

TC ID	Test Case	Input	Expected	Actual	Result
TC-09	GK 0 minutes = 0 pts	minutes=0	0	0	☒ PASS
TC-10	GK clean sheet, home	90 min, CS=1, venue=Home	6 pts	6	☒ PASS
TC-11	GK away clean sheet bonus	90 min, CS=1, venue=Away	8 pts	8	☒ PASS
TC-12	GK 6 saves (+bonus)	90 min, saves=6, no CS	6 pts	6	☒ PASS
TC-13	GK yellow card deduction	90 min, yellow=1	1 pt	1	☒ PASS
TC-14	GK red card deduction	90 min, red=1	-1 pt	-1	☒ PASS
TC-15	MID 1 goal	90 min, goals=1	7 pts	7	☒ PASS
TC-16	MID 1 assist	90 min, assists=1	5 pts	5	☒ PASS
TC-17	MID full game + high ICT	90 min, ICT=9.5	4 pts	4	☒ PASS

TC ID	Test Case	Input	Expected	Actual	Result
TC-18	MID null ICT (no crash)	ict_index=None	2 pts, no exception	2	 PASS
TC-19	FWD hat-trick	90 min, goals=3	14 pts	14	 PASS
TC-20	FWD high ICT, no goal	90 min, ICT=11.0	5 pts	5	 PASS
TC-21	Captain doubles score	MID 1 goal (7 base), captain=True	14 pts	14	 PASS
TC-22	Bench player = 0 pts	MID 2 goals, bench=True	0 pts	0	 PASS

UC3 – Leaderboard

TC ID	Test Case	Input	Expected	Actual	Result
TC-23	Singleton – same instance	GlobalLeaderboard() x2	lb1 is lb2	True	 PASS
TC-24	Team score stored	update_team_score(100, 'X', 45)	'X' in ranking	Found	 PASS
TC-25	Ranking sorted descending	Teams with score 10 and 99	99 first	99 first	 PASS
TC-26	Observer notified on update	Attach ConsoleNotifier, update score	'LEADERBOARD' in stdout	Logged	 PASS

UC4 – API Integration

TC ID	Test Case	Input	Expected	Actual	Result
TC-27	GET /players returns list	GET /players	HTTP 200, list	200 + list	 PASS
TC-28	GET /players?team=Arsenal	GET /players?team=Arsenal	All items team=Arsenal	Correct	 PASS
TC-29	GET /stats/latest	GET /stats/latest	HTTP 200	200	 PASS
TC-30	Sort param injection blocked	?sort_by=DROP_TABLE	Defaults to total_points	Defaulted	 PASS
TC-31	GET /player/1/history	GET /player/1/history	200 + player + history	Correct	 PASS
TC-32	GET /player/999999/history	GET /player/999999/history	HTTP 404	404	 PASS
TC-33	POST /team valid 15-player squad	15 valid players (2GK,5DEF,5MID,3FWD)	HTTP 200, team_id	200	 PASS
TC-34	POST /team non-existent player IDs	player_ids=[999991...]	HTTP 400	400	 PASS

Bug Fix Record

Bug ID	Description	Component	Root Cause	Fix
BUG-01	TypeError: '>=' not supported for NoneType on API startup	calculator.py	Stats columns return NULL for players not in a gameweek; NULL values passed to arithmetic comparisons.	Added a pre-processing loop in <code>calculate_team_score()</code> to replace None values with 0 before scoring logic.
BUG-02	fpl_fetcher.py saves to wrong database	fpl_fetcher.py	Hardcoded <code>sqlite3.connect('fantasy.db')</code> instead of using <code>core.database.get_db_connection()</code> .	Updated <code>save_to_db()</code> to resolve the DB path via the shared <code>core.database</code> module.
BUG-03	Database schema not created on first run	database_setup.py / api.py	<code>create_schema()</code> was not called before API startup.	Added <code>create_schema()</code> call inside <code>@app.on_event('startup')</code> in <code>api.py</code> .

Analysis & Observations

UC1 – Team Builder

The Strategy Pattern used for validation proved highly effective. Each rule (budget, size, position, club limit) is independently testable. A notable observation: the API router does not call `save_team()` before persisting to the database, so position and size validations are enforced at the object level only. This is a known trade-off logged as a potential improvement.

UC2 – Points Calculator

The scoring system correctly implements position-specific rules using the Strategy Pattern with a Decorator (ModifierStrategy) for captain and bench multipliers. Edge cases such as zero minutes, null ICT values, and away clean-sheet bonuses all behaved as expected after the BUG-01 fix. The None-handling fix was critical for stability during API startup.

UC3 – Leaderboard

The Singleton pattern was verified to return the same object reference across multiple instantiations. The Observer chain (leaderboard → EventBus → ConsoleNotifier / UIEventNotifier) fired correctly on every score update, demonstrating that cross-component communication does not require direct coupling between modules.

UC4 – API Integration

All REST endpoints responded correctly under nominal and edge conditions. The SQL injection prevention test confirmed that unsupported `sort_by` values are silently defaulted to `total_points`. The `/stats/latest` resolver correctly identified the highest gameweek, and the player history endpoint returned a proper 404 for unknown IDs.

Quality Metric Summary

Quality Factor	Metric	Target	Achieved
Correctness	% tests passed	100%	100%
Robustness	Crash on null input	0 crashes	0 crashes (post BUG-01 fix)
Security	SQL injection resistance	0 vulnerabilities	Passed
Modularity	Strategy pattern independence	Each rule independently testable	Achieved
Consistency	Singleton uniqueness	1 instance	Verified

Test Summary

Metric	Value
Total Test Cases	34

Metric	Value
UC1 – Team Builder	8
UC2 – Points Calculator	14
UC3 – Leaderboard	4
UC4 – API Integration	8
Passed	34
Failed	0
Bugs Found	3
Bugs Fixed	3